

human ability that is actually quite logical and mechanical, and that's where computers excel.

It's Already Happened

No, you can't go out and buy a self-driving car just yet, but there's already a race event for them!

The US DARPA (Defense Advanced Research Project Agency) has an event called the Grand Challenge for researchers to showcase their self-driving cars and pit them against each other in a race.

The first race took place in 2004. A hundred and six entries were vying for the prize—1 million US dollars. Quite a few crashed and burned, while others just got lost along the



Stanford's Stanley—smart, sturdy and unafraid of its own shadow

way. None of the entries were even able to finish the race, leave alone finish it in the stipulated time in order to win the prize money. Some veered off-course, trying to avoid colliding into their own shadows; some screeched to a halt for tiny bushes, but failed to recognize huge boulders... we may laugh, but the amount of money and time that was invested in this event was serious.

For the 2005 Grand Challenge, DARPA raised the stakes to \$2 million, and it worked! Stanford Racing Team's entrant, Stanley, safely completed the 132-mile desert course in just 6 hours and 54 minutes! Visit www.stanfordracing.org for more information on Stanley.

The 2007 Grand Challenge still gives away \$2 million, but the difference is that it will be held in more real-world scenarios—in a city, with other cars and traffic lights on the road. We'll keep you posted on how it turns out. The fact is that in the 2005 challenge, five cars were able to finish the race, and this is promising: these cars actually were "thinking" and taking "decisions" on their 132-mile trek, in off-road conditions! If even a single car can finish the 2007 race, we will be very tempted to believe that DARPA's goal of replacing one-third of cars with robotic driven vehicles might actually come true!

A Vision

If cars can already think well enough to navigate in off-road conditions with no prior mapping of terrain, imagine how good they'd be if the route were actually mapped! Is it that hard to imagine a day, maybe less than a decade from now, when you don't need to learn to whistle just to stop a cabbie in New York, or dodge recklessly-driven buses to cross the street in Mumbai?

Imagine just hopping into a car, stating your destination, then sitting back as the car decides by itself what the fastest route to your destination would be—factoring in information it receives via GPS about traffic conditions, accidents (if any), and also the shortest route... except for the fact that people might not like the idea of giving up control to a robot, it all sounds hunky-dory.

Another aspect of cars that might change—if adaptive guidance systems do become a reality—is design. Right now, cars are designed a certain way because things like driver comfort, visibility, aerodynamics, etc., all come into play. What happens when a driver isn't needed? Will we stick to the designs we have or will we change them radically?

Whatever happens, prepare to see some really cool technologies and designs in cars in the next 10 years. Until then, try not to choke too badly on your way to work... and dream on! ■

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